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Section B
11/24/2025

Data Collection and Descriptive Analysis

Data Collection Overview

Strategic Methodology

The purpose of this data collection is to gather a sufficient amount of data such that explanatory visuals could be derived. The original plan of the team was to incorporate at least five different tickers, but to aim to get 10 tickers. After further research, the team found that some of the tickers provided insufficient data. These tickers were discarded, and the 5-6 remaining tickers discussed were analyzed. Additionally, the team wanted to find data over the course of a 24-36 month period. An obstacle blocked this original goal, and the team had to settle for the analysis across a six-month interval beginning with January 1st, 2025, and ending on June 30th, 2025. Finally, the general data collection was done through the use of APIs to collect financial market data of each selected ticker and comments from Reddit posts in wallstreetbets subreddit page. The analysis and data collection took place using the programming language Python with the utilization of the web-based integrated development environment of JupyterLab for the development and construction of the corresponding code used to gather the data. Lastly, the team was aided by Generative AI for code development in order to optimize data collection. The critical thinking process and synthesis of the project ideas and skeptical breakdown of trends, interpretations, etc., reflect the thoughts and knowledge of the team, but some credit is due to Generative AI.

Dataset 1: Financial Market Data

We obtained structured market data from Yahoo Finance using the yfinance Python package for our target universe of Reddit Inc. (RDDT), NVIDIA Corporation (NVDA), Palantir Technologies (PLTR), Tesla Inc. (TSLA), Beyond Meat Inc. (BYND), GameStop Corp. (GME), and the Standard & Poor's (S&P) 500 index ETF SPY, which served as our market-control series. The available attributes for each ticker were Date, Open, High, Low, Close, Adjusted Close when available, and Volume. From these fields, we derived additional features such as daily returns, rolling moving averages, and short-window volatility. The main benefits of this data source were its reliability, standardized format across tickers, ease of programmatic retrieval, and readiness for time-series feature engineering. Key limitations involved the lack of consistent intraday data because yfinance provided daily price-and-volume data by default, occasional missing days for certain tickers, and the need to account for corporate actions when working with price data. Because our objective was to study sentiment-driven and non-valuation-based movements associated with wallstreetbets activity, we focused on price, volume, and volatility features rather than fundamental valuation metrics, and we used SPY to

control for broad market movements so we could isolate Reddit-specific effects. Data collection was implemented by looping over the ticker list with yfinance to download historical data, normalizing the date field, adding ticker identifiers, converting numeric fields, generating derived features, and saving the results. The final processed dataset was then uploaded into our project's MongoDB database as a clean, structured collection ready for integration with our Reddit sentiment data.

The financial dataset consisted of daily price and volume information for the six selected tickers over the full research window from January 1, 2025 through June 30, 2025 (6 months total). After importing and cleaning the raw data, the final collection contained just over 847 individual daily records, each storing fields such as date, ticker symbol, open, high, low, close, trading volume, and computed variables including daily return, a five-day moving average, and a five-day volatility measure. The structure of the documents in MongoDB showed that each observation corresponded to a single ticker-day pair, which provided a clean time-series format suitable for merging with the Reddit sentiment dataset and for running predictive analyses on the relationship between sentiment and subsequent stock performance.

Dataset 2: Reddit Unstructured Data

We collected unstructured social media data from Reddit using the PRAW Python client and the VADER sentiment analyzer from NLTK, targeting the wallstreetbets subreddit and a predefined list of tickers including SPY, RDDT, NVDA, PLTR, BYND, GME, and TSLA. For each post we extracted attributes such as post identifier, creation date, title, body text, combined full text (title + body text), detected tickers, post score, upvote ratio, number of comments, and four sentiment scores from VADER labeled positive, negative, neutral, and compound. The main benefits of this dataset were that it captured real-time retail investor discussion, provided both text and useful metadata for weighting and filtering posts, and allowed us to quantify community sentiment at a daily and ticker level. The primary limitations we encountered were that the data contained noise, slang, and sarcasm that complicated sentiment interpretation and the subreddit search approach returned an uneven number of posts across tickers. Data collection was automated with a script that authenticated to Reddit, iterated through each ticker with a search query limited to the target date range, combined post titles and bodies into a single text field, applied a regular expression to identify occurrences of our tickers, computed VADER sentiment scores for each post, and then aggregated the results by ticker and date to produce daily sentiment summaries suitable for merging with the structured stock data.

The Reddit sentiment data was collected from r/wallstreetbets using the PRAW API and focused on seven actively discussed tickers in retail trading communities, specifically SPY, RDDT, NVDA, PLTR, and TSLA. BYND and GME were part of the initial ticker list but were removed after returning little to no valid entries. The data collection process involved retrieving posts that referenced any of the selected tickers within the title or text and then applying VADER sentiment

analysis to compute daily average sentiment measures. After filtering and aggregating the posts within the target time frame, the final dataset contained 605 records. The sentiment data spanned from January 1, 2025 through June 30, 2025 and included daily averages for compound, positive, negative, and neutral sentiment, as well as average post score and average number of comments for each ticker.

Descriptive Analysis of Collected Data

The merged market-and-sentiment dataset contains 605 records across five tickers (NVDA, PLTR, RDDT, SPY, TSLA), each with 121 daily observations spanning January 1 to June 30, 2025. Each entry includes daily market features (Open, High, Low, Close, Volume, Return, MA_5, Vol_5) alongside Reddit-derived sentiment metrics (daily average compound, positive, negative, neutral scores, daily comment score, and the number of comments). To prepare the dataset, we cleaned and standardized the market data, computed daily returns and rolling volatility measures, aggregated Reddit comments to the daily level, calculated VADER-based sentiment averages, and aligned both sources by Ticker and Date. Summary statistics show that PLTR exhibited the highest mean return, while SPY showed the lowest volatility, consistent with its broad-market nature.

More analysis was acquired by examining the daily comment count. **Table 1** summarizes the daily comment count for each ticker that we acquired over the course of the six month interval that was investigated. The ticker with the highest acquired comment count was SPY with 3121 comments per day. Comparatively, the lowest daily comment count was discovered to be for PLTR ticker. Further exploratory analysis is demonstrated through the daily average compound sentiment.

Table 1. The daily count of comments for each ticker accumulated across the six months that were surveyed.

Ticker	**Daily Comment Count**
NVDA	1611
PLTR	414
RDDT	430
SPY	3121
TSLA	2166

Table 2. The tabular summary of the average daily compound sentiments observed for the ticker NVDA, PLTR, RDDT, SPY, and TSLA.

Ticker	daily_avg_compound
NVDA	-0.024853
PLTR	0.028738
RDDT	-0.130531
SPY	-0.032939
TSLA	-0.050434

To support later comparisons, **Table 2** reports the average daily compound sentiment score for each ticker. PLTR shows the highest average sentiment (.028738) across the six-month sample period, while every other ticker exhibits a negative average daily compound sentiment score. **Figure 1** further illustrates this by showing the distribution of daily compound sentiment across tickers, revealing notably wide distributions that indicate substantial day-to-day variation. On average, PLTR's sentiment is slightly positive, NVDA and SPY are neutral to slightly negative, and RDDT is strongly negative, recording the lowest sentiment of all tickers. Comment volume also varied sharply: SPY generated the most discussion activity (3,121 comments), followed by TSLA and NVDA. Overall, these patterns highlight the contrast between RDDT's consistently negative sentiment and PLTR's relatively positive tone.

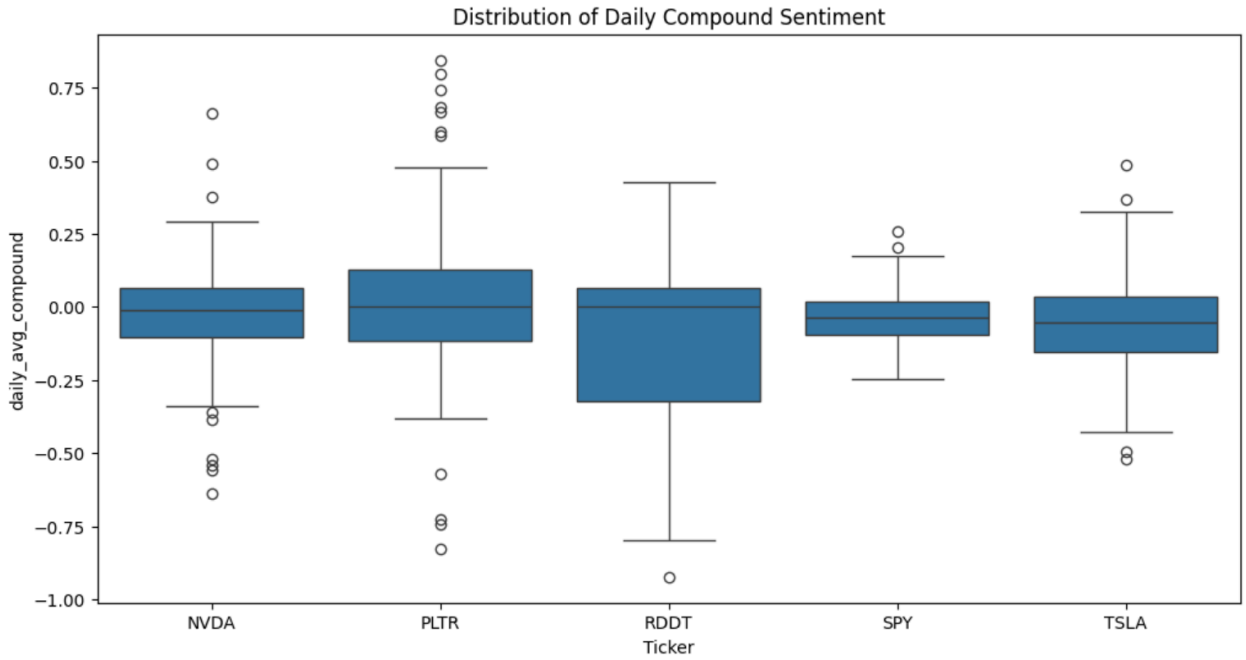

Figure 1. A boxplot of the distribution of daily compound sentiment acquired across tickers.

Further analysis was desired. This was done via the examination of the daily returns to the daily compound interest, which can be seen graphically in **Figure 2**. **Figure 2** overlays the different returns against the daily average compound sentiment for NVDA, PLTR, RDDT, SPY, and TSLA, respectively. Furthermore, this plot of daily sentiment against daily returns further illustrates the variability in sentiment–performance dynamics across tickers. Additionally, this conceptually motivates the predictive modeling in the next section. Together, these descriptive statistics and visuals provide a clear profile of both market behavior and sentiment activity within the merged sample.

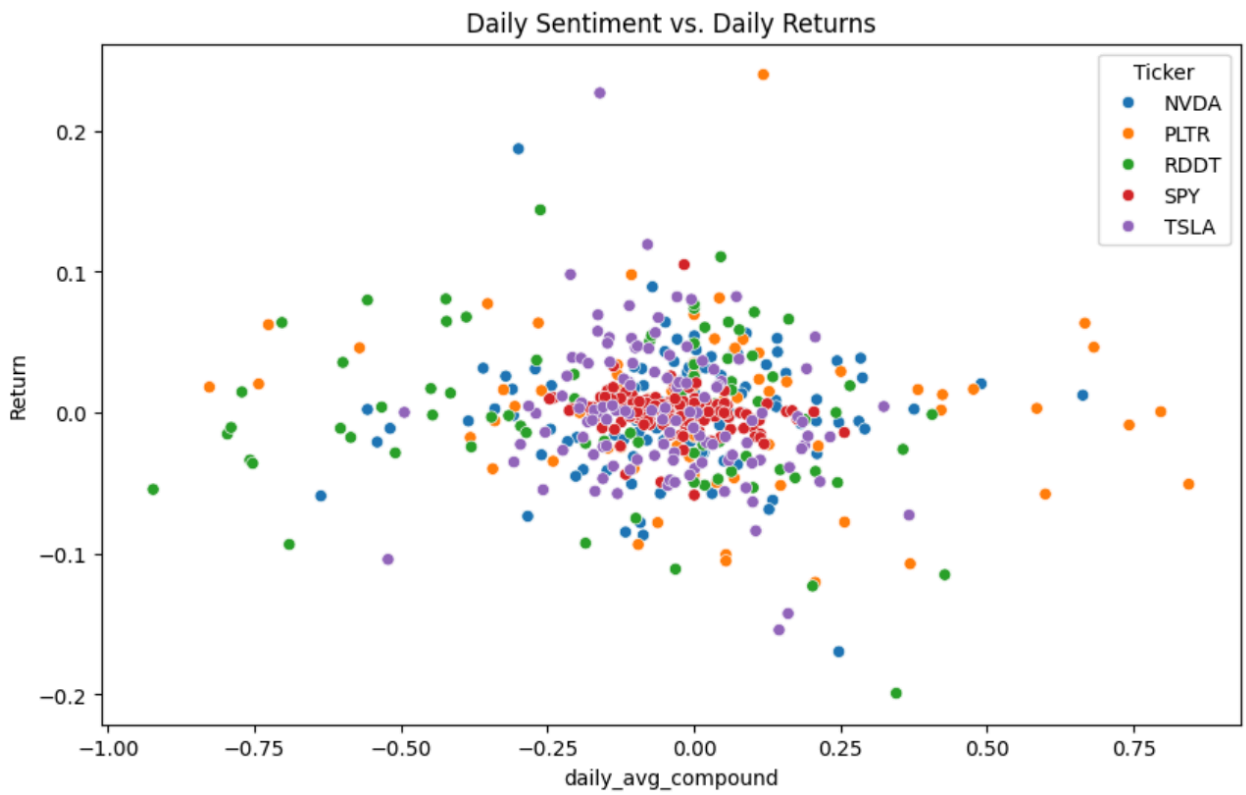

Figure 2. A scatterplot showing the relationship between daily compound sentiment and daily returns across the sample period for each ticker.

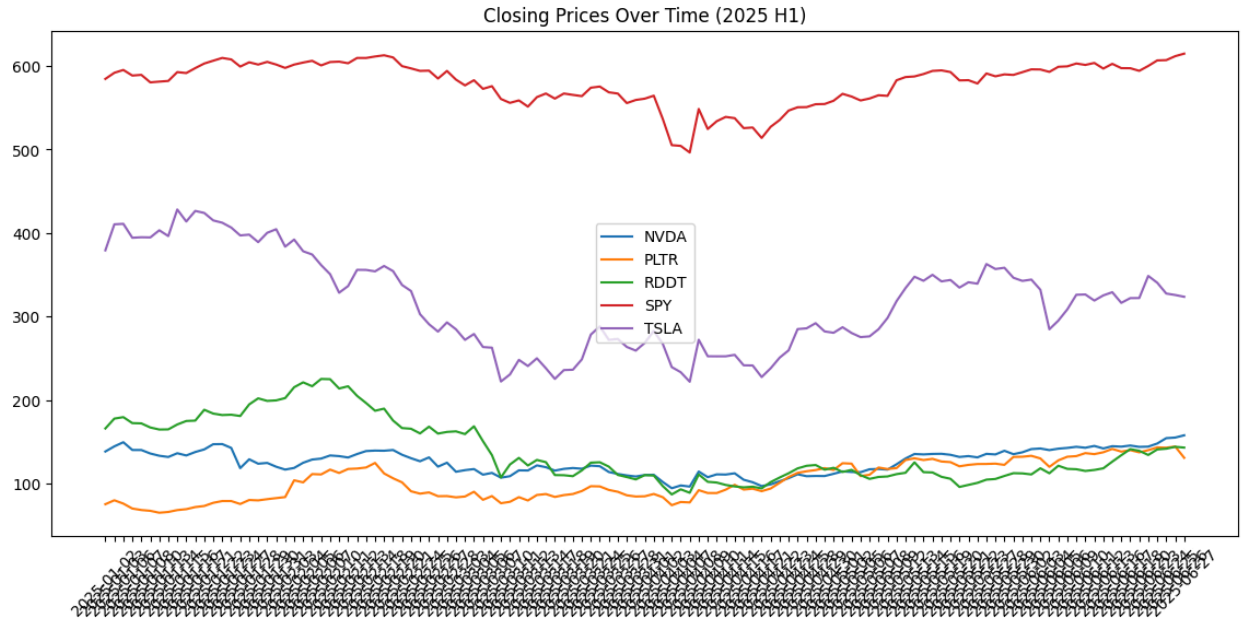

Figure 3. Daily closing prices for five equities across the first half of 2025.

The evolution of closing prices in the first half of 2025 was of interest to the team. Further investigation was done and data was gathered. A graphical visualization of this closing price time evolution can be seen in **Figure 3**. Some trends that are observed were: SPY showed high stability at elevated levels, TSLA exhibited a large mid-year decline followed by a recovery period, RDDT peaked early then declined significantly, and NVDA and PLTR showed steadier upward trends. Other insights that were immediately noticed were that distinct price levels indicated different risk profiles across equities, the divergence in the market behavior, and there was evidence of mid-year market disruptions. **Figure 3** also shows relative stability across different asset types, distinct behavior of AI/tech-oriented tickers (NVDA, PLTR) compared to high-variance growth stories, and the role of SPY as a benchmark index, showing minimal disruption relative to individual names. Also, **Figure 3** was helpful for contextualizing subsequent analyses such as volatility modeling, correlations between tickers, or sentiment-return dynamics.

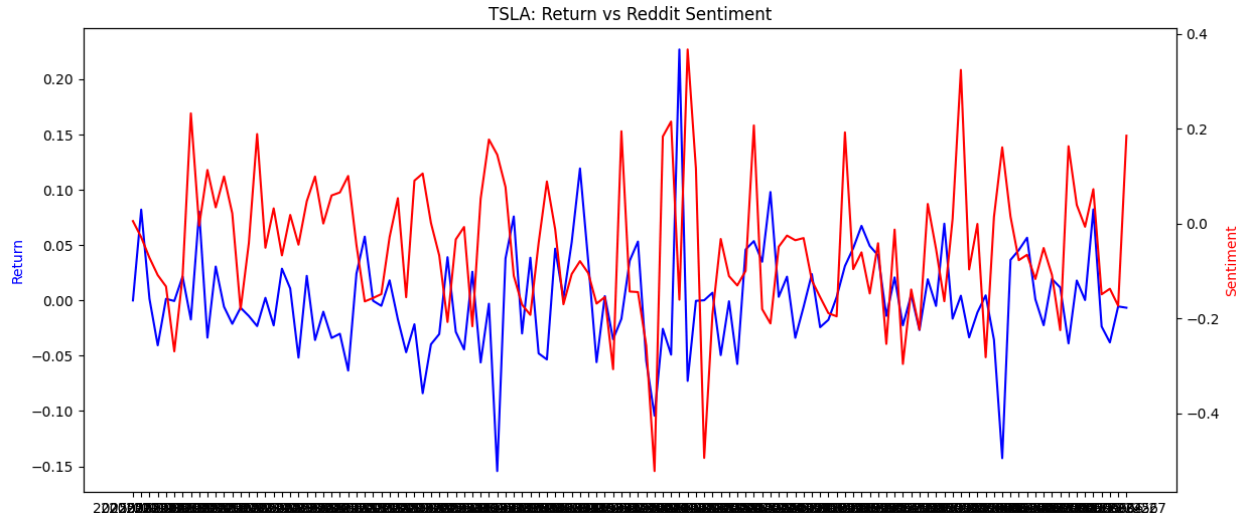

Figure 4. Time-series relationship between Tesla's daily stock returns and extracted daily Reddit sentiment, with the x-axis representing every day between the interval January 1st, 2025 to June 30th, 2025.

The TSLA return against Reddit sentiment showed some interesting trends. To better analyze these trends, the team graphed the time-series relationship between Tesla's daily stock returns and the corresponding daily sentiment extracted from Reddit discussions. **Figure 4** uses a dual-axis format to convey financial performance and sentiment signals measured on different numerical scales. Both series exhibit high day-to-day variability. The blue return line and red sentiment line both fluctuate sharply from day to day. This shows TSLA returns experience frequent reversals (positive and negative swings). Reddit sentiment is similarly volatile, reflecting rapid shifts in discussion tone. Also in **Figure 4**, neither variable displays a smooth or trending pattern across the horizon. The sentiment also appears more centered around zero. Specifically, the red sentiment line oscillates around the 0 mark throughout the entire time period. The significance of this oscillation observed in **Figure 4** reveals itself through the daily Reddit sentiment for TSLA being mixed. In other words, the daily Reddit sentiment is neither strongly positive, nor strongly negative. Some outlier days do appear with sentiment (sentiment > 0.20 or sentiment < -0.20), but are not frequent. The returns also show occasional extreme movements. The blue return line demonstrates several sharp positive spikes exceeding +0.15 and several negative drops of -0.10 to -0.15. These suggest periods of several high-volatility trading days.

The team took a different avenue and decided to explore the influence that the number of comments would have. The team decided to expand this idea to the five tickers of interest (NVDA, PLTR, RDDT, SPY, and TSLA). The result of the data collection can be shown by the trends observed in **Figure 5**. **Figure 5** displays the relationship between daily Reddit comment activity on selected stock tickers and their corresponding 5-day realized volatility. **Figure 5** displays a graphical visualization of the number of comments on a daily basis across a five-day

period. The scatter plot in **Figure 5** was decided on for the demonstration of this relationship. Each point in the chart represents a single trading day for one of the five equities NVDA, PLTR, RDDT, SPY, and TSLA. These tickers are differentiated by color and shown in the legend provided in **Figure 5**. The low comment days are the most frequent. Across all tickers, the majority of points cluster on the left side of the plot (between 0 and 10 comments per day). This is providing indication that most tickers do not consistently attract high Reddit activity and high comment events are relatively rare. Additionally, Most volatility values lie between 0.015 and 0.05. This suggests that the asset generally exhibits low to moderate short-term volatility and high volatility days (above 0.08-0.10) are comparatively uncommon.

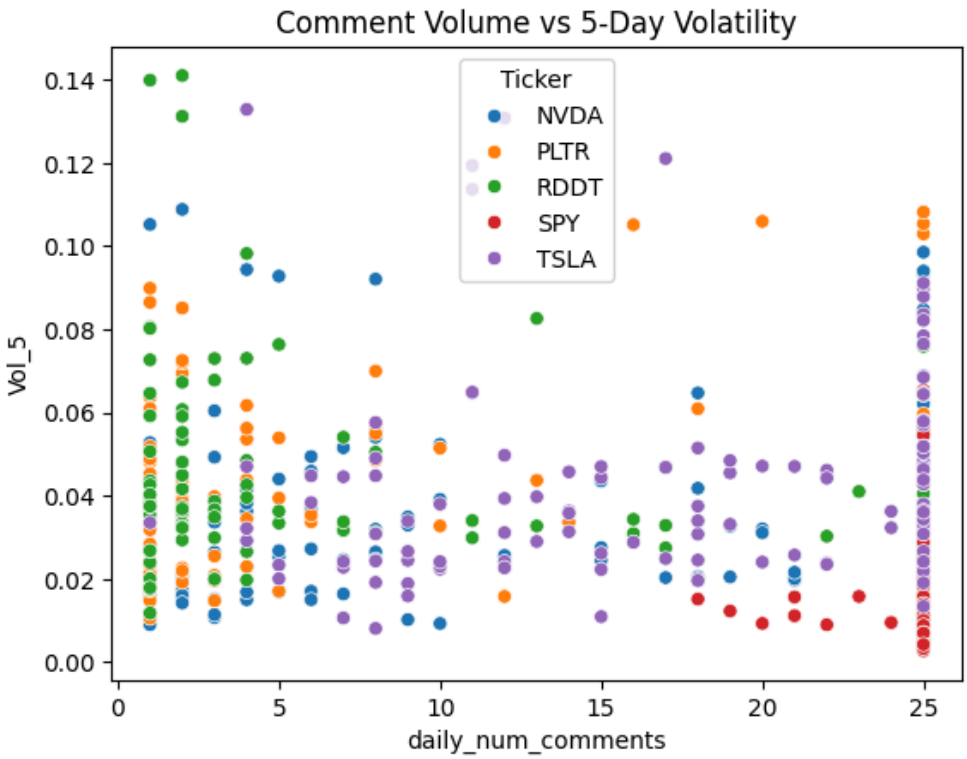

Figure 5. The volume of comments across a 5-day period in order to determine the volatility of the tickers NVDA (blue), PLTR (orange), RDDT (green), SPY (red), and TSLA (purple).

The daily average compound score can be compared with other sentiment scores. The positive and negative sentiments were investigated across the surveyed timeframe, and the corresponding daily averages of these sentiments were reported in **Table 3**. The highest average positive sentiment was found for the PLTR ticker (0.102271). The lowest average positive sentiment was found for the RDDT ticker (0.063020). The negative sentiment highest daily average value was for the RDDT ticker (0.126587). The lowest negative sentiment for a daily average belongs to the NVDA ticker. Furthermore, the summary statistics were also obtained and are shown in **Table 4**. The mean, standard deviation (STD), minimum value (MIN), and maximum value (MAX) were collected.

Ticker	daily_avg_compound	daily_avg_pos	daily_avg_neg
NVDA	-0.024853	0.080464	0.087685
PLTR	0.028738	0.102271	0.088354
RDDT	-0.130531	0.063020	0.126587
SPY	-0.032939	0.078509	0.091096
TSLA	-0.050434	0.087037	0.114853

Table 3. The daily average sentiment scores for the tickers NVDA, PLTR, RDDT, SPY, and TSLA.

Returns				
Ticker	Mean	STD	MIN	MAX
NVDA	0.001873	0.039683	-0.169682	0.187227
PLTR	0.005941	0.052887	-0.120465	0.239909
RDDT	0.000370	0.056772	-0.199209	0.247159
SPY	0.000546	0.016236	-0.058543	0.105019
TSLA	-0.000157	0.048520	-0.154262	0.226900

Table 4. A table of the summary statistics observed for the tickers NVDA, PLTR, RDDT, SPY, and TSLA.